A Two-Tier System for Remunerating Banks' Excess Reserves*

Paolo de Rosa†
CEMFI

26 July 2024

Abstract

This paper investigates the European Central Bank's (ECB) introduction of a two-tier system for remunerating banks' excess reserves, implemented in 2019. The tiering system exempted a fraction of banks' excess reserves from negative remuneration to mitigate the adverse effects of prolonged negative policy rates. Despite its significance, the economic literature has yet to examine this measure in detail. This study develops a macroeconomic model to analyze the impact of tiering on the transmission of monetary policy. The findings suggest that tiering facilitates the transmission of expansionary monetary policy by increasing banks' equity and relaxing capital constraints, thereby boosting lending and economic output while also raising the inflation rate. However, the associated costs reduce household consumption, highlighting a trade-off in the policy's effectiveness. The results underscore tiering as a viable unconventional monetary policy tool during periods of negative policy rates, with implications for future central bank strategies.

*I am grateful to Rafael Repullo for his support and guidance at different stages of the project. I thank Vedant Agarwal, Toni Ahnert, Martín Farias, Jan Lukas Schäfer, Enrique Sentana, and Javier Suarez for their comments and suggestions, as well as seminar participants at CEMFI. I gratefully acknowledge financial support from Grant PRE2022-000868 funded by MCIN/AEI/10.13039/501100011033 and by "ESF+", the Maria de Maeztu Unit of Excellence CEMFI MDM-2016-0684 and CEMFI CEX2020-001104-M, funded by MCIN/AEI/10.13039/501100011033, and CEMFI.

†E-mail: paolo.derosa@cemfi.edu.es.

1 Introduction

In 2019, the European Central Bank introduced a two-tier system for remunerating banks' excess reserves (tiering), which consisted of an exemption from negative remuneration for a fraction of banks' excess reserves. The measure was intended to limit any undesired consequences of the ECB's long negative policy rate. However, since 2019, the economic literature has not studied the measure in detail. I aim to bridge this gap by developing a macroeconomic model to study the effect of tiering on the transmission of monetary policy. I find that tiering indeed helps with the transmission of an expansionary monetary policy.

In 2014, in an attempt to stimulate the economy, the ECB started remunerating banks' excess reserves at a negative rate, setting a negative deposit facility rate (DFR). In the absence of cash, banks could simply pass those negative rates to their depositors. However, this was not possible because, at least in theory, depositors could massively shift to cash if offered a negative deposit rate. In other words, there probably exists a zero lower bound on deposit rates that banks cannot cross if they want to maintain the commercial relationship with their depositors.

As shown in Figure 1, banks never crossed such a lower bound and continued offering a slightly positive interest rate to their depositors even when the policy rate was negative. This meant that banks were actually making losses on deposits, at least at the margin, since they were paying non-negative interests on deposits and investing the relevant funds in reserves that paid negative rates.

The reason why banks accepted this situation is probably that they wanted to maintain the commercial relationship with their depositors, hoping that, sooner or later, policy rates would return to normal. However, this choice reduced their profitability. While central banks do not have banks' profitability as an objective, over time, reduced profitability in the banking sector may have a negative impact on lending via capital requirements (Abadi et al., 2023) and impair the transmission of an expansive monetary policy.

Therefore, to alleviate possible negative consequences of its negative interest rate policy (Draghi and Guindos, 2019), in October 2019, the ECB introduced a two-tier system for remunerating banks' excess reserves. The measure was designed as follows: excess reserves

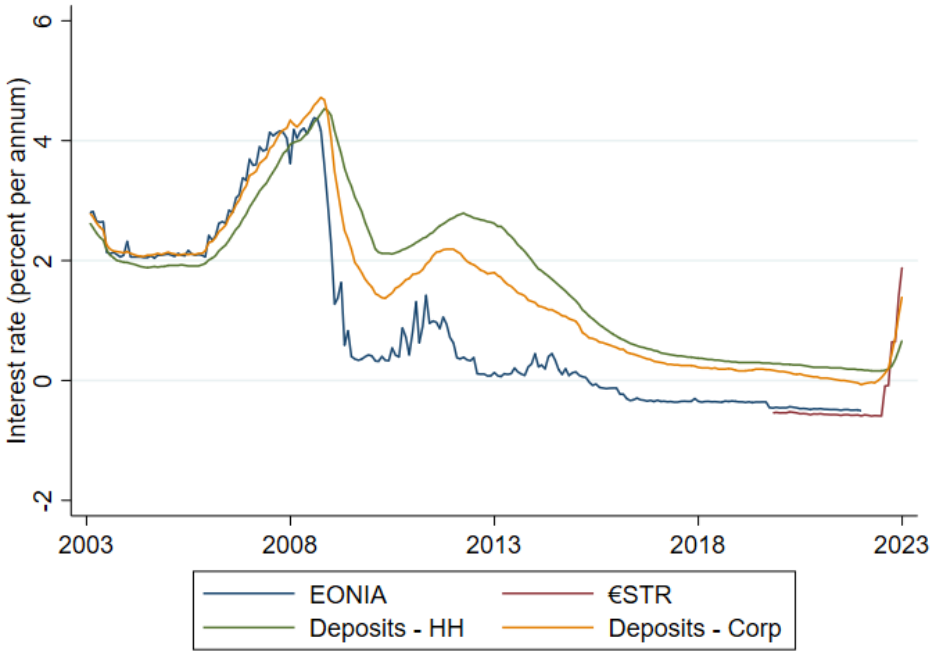


Figure 1: Interest Rate on Deposits versus the Policy Rate in the Eurosystem

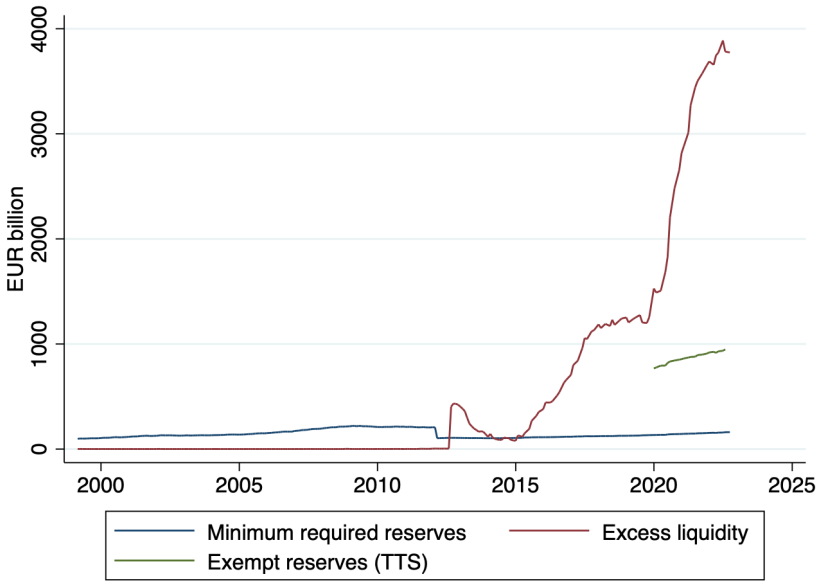
up to six times the minimum required reserves (1% of specific liabilities, mainly deposits) were remunerated at zero (i.e., they were exempted from negative remuneration), while the remaining excess reserves continued to be remunerated at the (negative) DFR.

The size of the exemption was calibrated to be significantly smaller than the total amount of reserves, so as not to alter banks' decisions at the margin, and the small number of banks that fell short of the exemption reacted by immediately borrowing reserves in the interbank market to fill it up (Boucinha et al., 2022). Therefore, from the point of view of banks, tiering was equivalent to a transfer of:

$$6\%\text{Deposits} \times (0 - \text{DFR}).$$

Given that the amount of exempt reserves was on average 850 billion euros and considering a DFR of -0.50 (as in 2019), this implied a transfer from the Eurosystem to banks of about 4.25 billion euros annually. Arguably, the amount of the transfer was relatively small relative to the size of the Eurosystem. Nonetheless, tiering deserves attention for

Figure 2: Reserves



its potential on a larger scale.

Finally, in September 2022, when the DFR became non-negative again, tiering was terminated.

Figure 2 shows the evolution of required and excess reserves and the amount of exempted reserves under tiering.

The literature on tiering is quite limited. An interesting work by Boucinha et al. (2022) reports that tiering was effective in increasing banks' profits without creating distortions; that it increased activity in the money market without affecting interbank rates; and that it was interpreted as a signal that negative interest rates would stay for long. A later study by Altavilla, Boucinha, Burlon, et al. (2022) finds empirical evidence that tiering increased lending (in quantities). An unpublished study by Eisenschmidt et al. (2022) finds limited empirical evidence that tiering increased deposit rates.

This literature, however, is not explicitly grounded in a complete theoretical model, and therefore, arguably lacks a theoretical foundation. With this work, I aim to fill this gap by developing and solving a Neo-Keynesian model with a financial sector, similar to that of Abadi et al. (2023), that allows the study of the effect of tiering.

In my model, I find that, after a large monetary policy shock, by increasing banks' equity and relaxing capital constraints, tiering increases lending and reduces lending rates, as well as increasing output and the inflation rate. However, due to its cost, tiering reduces consumption. This points towards the conclusion that tiering is indeed an effective unconventional monetary policy tool to stimulate the economy when policy rates are negative.

2 Model Economy

I consider a New-Keynesian economy in discrete time $t \in \{0, 1, 2, \dots\}$, similar to that of Abadi et al. (2023). I introduce some changes in the way banks and the central bank are modeled that allow the study of tiering.

The economy is populated by households that work, consume, save, and are entitled to all profits. Production is delegated to four types of firms: intermediate goods producers that use labor and capital to produce intermediate inputs; a continuum $k \in [0, 1]$ of monopolistic retailers that differentiate and resell those inputs; final goods producers that produce consumption goods by assembling differentiated varieties; and capital goods producers that transform consumption goods into capital. There is a continuum $j \in [0, 1]$ of monopolistically competitive banks that take deposits to invest in loans, bonds, and reserves. Finally, there is a central bank that supplies reserves and implements monetary policy following a Taylor rule, and a government that issues long-term bonds and sets taxes and transfers.

The next subsections describe all agents in detail. Detailed derivations are left for Appendix A.

2.1 Households

Households choose consumption C_t, labor supply H_t, and allocate savings between one-period safe bonds B_t^h and safe deposits $D_{j,t}$ to maximize their lifetime utility:

$$\max_{\{C_t,H_t,B_t^h,\{D_{j,t}\}_{j\in[0,1]}\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(\frac{(C_t - hC_{t-1})^{1-\sigma}}{1-\sigma} - \chi \frac{H_t^{1+\phi}}{1+\phi} + \zeta \log \left(\int_0^1 D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj \right)^{\frac{\epsilon^D}{\epsilon^D - 1}} \right) \right], \quad (1)$$

where β is the discount factor, σ is the elasticity of intertemporal substitution, h is the scale parameter of habit formation, χ is the scale factor of the disutility of labor, ϕ is the Frisch elasticity of labor supply, and ζ is the scale factor of the utility provided by holding deposits.

Following the *money in the utility function* tradition (dating back to Feenstra, 1986 and Poterba and Rotemberg, 1986), I assume that holding deposits directly enters the utility function. In addition, following Gerali et al. (2010), I assume constant elasticity of substitution among deposits of different banks[1]. This assumption allows us to replicate the negative spread between deposits and policy rates we observe in aggregate data for normal times.

Households take nominal wages W_t, the price level P_t, and rates on bonds $1 + i_t$ (which corresponds to the policy rate) and deposits $1 + i_{j,t}^D$ as given. Therefore, their (real terms) budget constraint is:

$$C_t + B_t^h + \int_0^1 D_{j,t} dj \leq \frac{W_t}{P_t} H_t + \frac{1 + i_{t-1}}{1 + \pi_t} B_{t-1}^h + \int_0^1 \frac{1 + i_{j,t-1}^D}{1 + \pi_t} D_{j,t-1} dj + \Pi_t^f + \Pi_t^b + T_t, \quad (2)$$

where $1 + \pi_t = \frac{P_t}{P_{t-1}}$ is inflation, Π_t^f and Π_t^b denote net dividends paid out by firms and banks, and T_t are lump-sum transfers from the government.

[1]This assumption can be interpreted as different members of the representative households preferring different banks. Given that banks are symmetric, in equilibrium all banks will supply the same quantity at the same rate. This is particularly convenient because the CES aggregator coincides with the actual amount of deposits supplied on aggregate, making interpretation easier. I.e.: $\int_0^1 (D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj)^{\frac{\epsilon^D}{\epsilon^D - 1}} = \int_0^1 D_{j,t} dj$.

Finally, I assume that households cannot short bonds or deposits: $B_t^h, D_t \geq 0$.

2.2 Intermediate goods producers

Intermediate goods producers are set up with no own funds and operate in two periods. In period $t-1$, they borrow to purchase capital K_t. In period t, they use capital and hire labor H_t to produce intermediate goods X_t, sell output at competitive price P_t^X, sell back undepreciated capital, repay debt, pay out dividends, and shut down.

Following Abadi et al. (2023), I assume there exist two types of intermediate goods producers: a mass ξ, indicated by subscript b, which is bank-dependent and relies on bank loans for financing capital, and a mass $(1-\xi)$, indicated by subscript n, which is non-bank-dependent and can issue one-period bonds at the policy rate.

Producers of type z have a neoclassical production function:

$$X_{z,t} = A_z(K_{z,t}^{\alpha} H_{z,t}^{1-\alpha})^{\nu}, \tag{3}$$

where A_z is type-specific total factor productivity, $K_{z,t}$ and $H_{z,t}$ are, respectively, capital and labor employed by type z, α is the share of capital, and ν determines the return to scale. I assume diminishing returns to scale, i.e., $\nu \in (0,1)$, so that intermediate goods producers of both types produce in equilibrium.[2]

The representative bank-dependent intermediate goods producer maximizes:

$$\max_{K_{b,t}, H_{b,t}} P_t^X A_b(K_{b,t}^{\alpha} H_{b,t}^{1-\alpha})^{\nu} + (1-\delta) P_t Q_{b,t}^K K_{b,t} - \int_0^1 (1 + i_{j,t-1}^L) L_{j,t-1} dj - W_t H_{b,t} \tag{4}$$

where $1-\delta$ is the depreciation rate of capital, $Q_{b,t}^K$ is the competitive real price of capital employed by bank-dependent firms, and $1 + i_{j,t}^L$ is the nominal rate on bank loans $L_{j,t}$ supplied by bank j (taken as given).

Following Gerali et al. (2010), I assume constant elasticity of substitution among loans of different banks[3]. This assumption allows us to replicate the positive spread between

[2]Under constant or increasing returns to scale, the less efficient bank-dependent intermediate goods producers would not produce anything.

[3]This assumption can be interpreted as different bank-dependent firms preferring different banks. Given that banks are symmetric, in equilibrium they will supply the same quantity at the same rate.

loan and policy rates we observe in aggregate data. Therefore, to finance capital, bank-dependent intermediate goods need:

$$\left(\int_0^1 L_{j,t-1}^{\frac{\epsilon^L-1}{\epsilon^L}} dj\right)^{\frac{\epsilon^L}{\epsilon^L-1}} \geq P_{t-1} Q_{b,t-1}^K K_{b,t}, \tag{5}$$

On the other hand, non-bank-dependent intermediate goods producers maximize the simpler:

$$\max_{K_{n,t},H_{n,t}} P_t^X A_n (K_{n,t}^\alpha H_{n,t}^{1-\alpha})^\nu + (1-\delta) P_t Q_{n,t}^K K_{n,t} - (1+i_{t-1}) P_{t-1} Q_{n,t-1}^K K_{n,t} - W_t H_{n,t}, \tag{6}$$

where $Q_{n,t}^K$ is the competitive real price of capital employed by non-bank-dependent firms, and $1+i_t$ is the policy rate.

Both types of intermediate goods producers take demand from retailers as given.

2.3 Retailers

A continuum $k \in [0,1]$ of retailers purchase intermediate goods at a competitive price and costlessly differentiate them under the production function:

$$Y_{k,t} = X_{k,t}. \tag{7}$$

where $Y_{k,t}$ is the quantity of differentiated goods produced by retailer k and $X_{k,t}$ is the quantity of intermediate goods it entails.

Retailers then sell their output to final goods producers at price $P_{k,t}$ under monopolistic competition and subject to Rotemberg (1982) price adjustment costs.

Each retailer maximizes:

$$\max_{\{P_{k,t}\}_{t=0}^\infty} \mathbb{E}_0 \left[\sum_{t=0}^\infty \beta^t \Lambda_t \left[P_{k,t} X_{k,t} - P_t^X X_{k,t} - \theta \left(\frac{P_{k,t}}{P_{k,t-1}} - 1\right)^2 Y_t\right]\right], \tag{8}$$

This is particularly convenient because the CES aggregator coincides with the actual amount of loans supplied on aggregate, making interpretation easier. I.e.: $\int_0^1 (L_{j,t}^{\frac{\epsilon^L-1}{\epsilon^L}} dj)^{\frac{\epsilon^L}{\epsilon^L-1}} = \int_0^1 L_{j,t} dj$.

where Λ_t denotes the households' marginal utility of consumption and θ is the scale parameter of the price adjustment cost. Taking demand from final goods producers as given.

2.4 Final goods producers

Final goods producers aggregate differentiated varieties supplied by monopolistic retailers to produce output under the production function:

$$Y_t = \left[\int_0^1 (Y_{k,t})^{\frac{\epsilon-1}{\epsilon}} dk\right]^{\frac{\epsilon}{1-\epsilon}}, \tag{9}$$

where ϵ is the elasticity of substitution among varieties. They then sell at a competitive price P_t.

So, the representative final good producer maximizes:

$$\max_{\{Y_{k,t}\}_{k\in[0,1]}} P_t \left[\int_0^1 (Y_{k,t})^{\frac{\epsilon-1}{\epsilon}} dk\right]^{\frac{\epsilon}{1-\epsilon}} - \int_0^1 P_{k,t} Y_{k,t} dk, \tag{10}$$

taking demand for final goods as given.

As in standard CES problems, final goods producers demand:

$$Y_{k,t}(P_{k,t}) = \left(\frac{P_{k,t}}{P_t}\right)^{-\epsilon} Y_t, \tag{11}$$

and competition drives the price of final goods to:

$$P_t = \left(\int_0^1 (P_{k,t})^{1-\epsilon} dk\right)^{\frac{1}{1-\epsilon}}. \tag{12}$$

which coincides with the price level we defined above.

2.5 Capital goods producers

Following Abadi et al. (2023), I assume there are two types $z \in \{b, n\}$ of capital goods producers that use final goods to produce capital employed by bank-dependent and non-

bank-dependent intermediate firms, respectively.

There are investment adjustment costs parameterized by κ^I, and the capital accumulation equation reads:

$$K_{z,t+1} = (1-\delta)K_{z,t} + I_{z,t+1}\left[1 - \frac{\kappa^I}{2}\left(\frac{I_{z,t+1}}{I_{z,t}} - 1\right)^2\right], \tag{13}$$

where $I_{z,t}$ denotes investment in sector z.

So, capital goods producers of type z maximize:

$$\max_{\{I_{z,t}\}_{t=0}^{\infty}} \mathbb{E}_0\left[\sum_{t=0}^{\infty}\beta^t\Lambda_t\left\{Q_{z,t}^K I_{z,t+1}\left[1 - \frac{\kappa^I}{2}\left(\frac{I_{z,t+1}}{I_{z,t}} - 1\right)^2\right] - I_{z,t+1}\right\}\right], \tag{14}$$

taking demand for capital as given.

2.6 Banks

A continuum $j \in [0,1]$ of infinitely-lived banks raise net worth $N_{j,t}$ and deposits $D_{j,t}$ to extend loans $L_{j,t}$, purchase corporate bonds $B_{j,t}^b$, long-term government bonds $G_{j,t}^b$, and invest in central bank reserves $R_{j,t}$, under the balance sheet identity:

$$N_{j,t} + D_{j,t} = L_{j,t} + B_{j,t}^b + Q_t^G G_{j,t}^b + R_{j,t}. \tag{15}$$

where Q_t^G is the price of government bonds. Since banks are fully symmetric, in what follows, I suppress the subscript j.

Banks are subject to a *public service constraint* such that they must always offer a non-negative rate on deposits $i_t^D \geq 0$ and accommodate the corresponding demand up to threshold D^*. It is possible to micro-found this constraint by assuming that households have two types of members: "depositors" who can invest in liquid assets and "savers" who cannot.

In addition, banks' lending entails a leverage cost $\Psi(N_t, L_t)$ that is decreasing in net worth

and increasing in lending. Following Abadi et al. (2023), I assume:

$$\frac{\partial \Psi(N_t, L_t)}{\partial L_t} = \kappa^L \left(\max \left\{ \frac{L_t}{N_t} - \frac{L_{SS}}{N_{SS}}; 0 \right\} \right)^2, \tag{16}$$

where κ^L is a scale parameter, and L_{SS} and N_{SS} are, respectively, lending and banks' net worth in steady state. The cost is a smooth approximation of the typical net worth constraints (e.g., Kiyotaki and Moore, 1997, Bernanke et al., 1999, or Mendicino et al., 2020). Using a smooth instead of a sharp constraint allows replication of the typical net worth buffers we observe in real-world data.

However, frictions in the bank equity market prevent banks from picking net worth optimally. Following Abadi et al. (2023), I model such frictions as follows: In each period, banks pay out a fixed fraction γ of their net worth as dividends and issue a constant amount of new equity $\hat{N}$. Therefore, net worth evolves as:

$$N_{t+1} = (1-\gamma) \left(\frac{1+i_t}{1+\pi_{t+1}} (B_t^b + Q_t^G G_t^b + R_t) + \frac{1+i_t^L}{1+\pi_{t+1}} L_t \right. \\ \left. - \frac{1+i_t^D}{1+\pi_{t+1}} D_t - \Psi(N_t, L_t) + CG_t^b \right) + \hat{N}, \tag{17}$$

where CG_t^b are capital gains of banks.

Government bonds have a face value of 1 and mature with probability $\frac{1}{\tau}$. Thus, no arbitrage implies that their real price Q_t^G solves:

$$\frac{1+i_t}{1+\pi_{t+1}} Q_t^G = \left(1 - \frac{1}{\tau}\right) Q_{t+1}^G + \frac{1}{\tau} \frac{1}{1+\pi_{t+1}}. \tag{18}$$

The return on government bonds must equate the risk-free rate. The holding of long-term bonds implies that banks have a maturity mismatch. Therefore, after a shock that unexpectedly changes the policy rate, the price of government bonds changes and banks make capital gains:

$$CG_t^b = \left(1 - \frac{1}{\tau}\right) \left(Q_t^G - \mathbb{E}_{t-1}[Q_t^G]\right) G_{t-1}^b. \tag{19}$$

Banks' objective is maximizing the discounted stream of dividends:

$$\max_{\{B_t^{lt}, i_t^L, i_t^D\}_{t=0}^{\infty}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \Lambda_t \gamma N_t \right], \tag{20}$$

taking demand for loans and deposits as given.

The complete solution of banks' problem is in Appendix A. However, let me anticipate that equilibrium rates are:

$$1 + i_t^L = \frac{\epsilon^L}{\epsilon^L - 1} \left(1 + i_t + \frac{\partial \Psi(N_t, L_t)}{\partial L_t} \right), \tag{21}$$

$$1 + i_t^D = \max \left\{ \frac{\epsilon^D}{\epsilon^D - 1} \left(1 + i_t \right); 1 \right\}, \tag{22}$$

which (except for the max operator in the deposit rate, which is due to the public service constraint) is the typical solution of monopolistic competition problems:

$$\frac{elasticity}{elasticity - 1} \times \text{marginal cost.}$$

2.7 Central bank

The central bank sets the nominal rate $1 + i_t$ according to a conventional Taylor rule with inertia:

$$\frac{1 + i_t}{1 + i_{SS}} = \left(\frac{1 + i_{t-1}}{1 + i_{SS}} \right)^{\rho^{mp}} \left(\frac{1 + \pi_t}{1 + \pi_{SS}} \right)^{\phi^{\pi}(1-\rho^{mp})} \epsilon_t^{mp} \tag{23}$$

where $1 + i_{SS}$ and $1 + \pi_{SS}$ are, respectively, the policy rate and inflation of steady state, ρ^{mp} is the persistence of the nominal rate, ϕ^{π} is the Taylor rule coefficient, and ϵ_t^{mp} is a monetary policy shock.

The central bank also operates the balance sheet:

$$R_t = G_t^c, \tag{24}$$

where R_t are reserves and G_t^c are government bonds held by the central bank. The central bank can decide the amount of reserves autonomously and sets it to $R_t = R^*$.

Similarly to banks, the maturity mismatch in the central bank balance sheet implies that the central bank makes capital gains:

$$CG_t^c = \left(1 - \frac{1}{\tau}\right)\left(Q_t^G - \mathbb{E}_{t-1}[Q_t^G]\right) G_{t-1}^c, \tag{25}$$

after an unexpected change in the policy rate.

Finally, the central bank can set the parameter η of a two-tier system for remunerating reserves, such that reserves up to η times deposits are remunerated at zero when the policy rate is negative. I assume $0 \leq \eta < \frac{R_t}{D_t}$ so that banks always fill the exemption entirely and their marginal investment opportunity is at the policy rate.

Therefore, the central bank makes profits:

$$\Pi_t^c = \begin{cases} CG_t^c & \text{if } i_t \geq 0 \\ CG_t^c + i_t \eta D_t & \text{if } i_t < 0 \end{cases}, \tag{26}$$

which are rebated to the government. Notice that $i_t \eta D_t$ is negative if $i_t < 0$.

From the point of view of banks, tiering is tantamount to a transfer $-i_t \eta D_t$ received from the central bank when the policy rate is negative.

2.8 Government

The government elastically supplies long-term bonds $G_t = G_t^b + G_t^c$ at price Q_t^G, invests the proceedings in an international asset A_t that perfectly maintains real value over time, and raises taxes T_t to service the debt.

Its budget constraint is:

$$A_t - Q_t^G G_t = A_{t-1} - \frac{1 + i_{t-1}}{1 + \pi_t} Q_{t-1}^G G_{t-1} + T_t - (CG_t^b + CG_t^c) + \Pi_t^c. \tag{27}$$

Since the government wishes to maintain $Q_t^G G_t = A_t$ at all times, it covers all interest expenditure and capital losses/gains net of transfers from the central bank with taxes.

$$T_t = \frac{i_{t-1} - \pi_t}{1 + \pi_t} Q^G_{t-1} G_{t-1} + (CG^b_t + CG^c_t) - \Pi^c_t. \tag{28}$$

3 Calibration

I calibrate the model to the Euro area, where tiering was first introduced. I set the length of a period to one quarter and follow the calibration strategy of Abadi et al. (2023) for comparability.

Households: I set the time rate of preference $\beta = 0.995$ to match a yearly real interest rate of 2%. The intertemporal elasticity of substitution $\sigma = 1$ and scale parameter of habit formation $h = 0.62$ follow the conventional values in the DSGE literature. Setting the scale parameter of the disutility of labor $\chi = 0.41$ is a normalization so that households work a quarter of their available time in the economy's steady state. The inverse Frisch elasticity $\phi = 2$ is, again, standard in the DSGE literature. The scale parameter of the liquidity demand $\zeta = 0.0045$ is described below in the paragraph about banks. Table 1 summarizes households' parameters.

Table 1: Households' parameters

Parameter	Description	Value	Target
β	Time rate of preference	0.995	Real rate
σ	Intertemporal elasticity	1	DSGE literature
h	Habit formation	0.62	DSGE literature
χ	Disutility of labor	0.41	Labor hours
ϕ	Inverse Frisch elasticity	2	DSGE literature
ζ	Deposits demand	0.0045	Loan/bonds ratio

Intermediate goods producers: I set the capital share $\alpha = 0.36$ following the convention in the DSGE literature. The returns to scale parameter $\nu = 0.85$ is set to match a steady state investment-consumption ratio of 2.7.

I identify bank-dependent firms with small and medium enterprises. In Eurostat data (Abadi et al., 2023), they constitute 99.8% of the total firms and account for 55.8% of output. Therefore, I set the fraction of bank-dependent firms $\xi = 0.998$, and the ratio

of TFPs $\frac{A^b}{A^n} = 0.43$ to match these two features of the data. I then set the TFP of non-bank-dependent firms $A^n = 1$ as a normalization.

Retailers: I set the retail price elasticity $\epsilon = 3.9$ and the scale parameter of the Rotemberg (1982) price adjustment cost $\theta = 70.7$ following the convention in the DSGE literature.

Capital goods producers: I set capital depreciation $\delta = 0.025$ following the DSGE literature ($\approx 10\%$ annual depreciation). The scale parameter of capital adjustment cost $\kappa^I = 5$ is set so that the elasticity of investment to a shock to the price of capital $\frac{1}{\kappa^I}$ is 0.2, as estimated by Smets and Wouters (2003) for the Euro area. Table 2 summarizes firms' parameters.

Table 2: Firms' parameters

Parameter	Description	Value	Target
α	Capital share	0.36	DSGE literature
ν	Returns to scale parameter	0.85	Consumption/investment ratio
ξ	Fraction of bank-dependent firms	0.998	Fraction of SMEs
A^b	TFP of bank-dependent firms	0.43	SME output share
A^n	TFP of non-bank-dependent firms	1	Normalization
ϵ	Retail price elasticity	3.9	DSGE literature
θ	Price adjustment cost	70.7	DSGE literature
δ	Capital depreciation	0.025	DSGE literature
κ^I	Capital adjustment cost	5	Elasticity of I_t to Q_t^I

Banks: I set the elasticity of loan demand $\epsilon^L = 200$ to target a spread of 2% in steady state between the loan and the policy rate, as estimated in Freriks and Kakes (2021), and $\epsilon^D = -275$ to target a steady-state spread of 1% between the policy and the deposit rate, as in Abadi et al. (2023).

Banks' dividend payout ratio $\gamma = 0.06$, the deposit demand scale parameter $\xi = 0.0045$, and the maturity of long-term bonds $\tau = 13.6$ are set to match three targets: a steady-state net worth-to-loan ratio of 0.155 (corresponding to an average Tier-1 capitalization ratio of 15.5% as in Altavilla, Boucinha, and Peydró, 2018), a steady-state loan-to-bond

ratio on bank balance sheets of 3.6, and an average maturity of bank bond holdings of 3.4 years (both reported by Hoffmann et al., 2019).

I set the threshold of the deposits public service constraint on deposits to $D^* = 6.93$ in order to match the increase in the deposit-to-GDP ratio reported in Abadi et al. (2023) from 2000 until 2014, when interest rates went negative for the first time. Finally, the scale parameter of the capital adjustment cost $\kappa^L = 0.018$ is set to replicate an elasticity of the interest rate on loans i_t^L to the target capitalization ratio $\frac{L^{SS}}{N^{SS}}$ of 28 basis points, as estimated by Macroeconomic Assessment Group (2010). Table 3 summarizes banks' parameters.

Table 3: Banks' parameters

Parameter	Description	Value	Target
ϵ^L	Elasticity of loan demand	200	Loan spread
ϵ^D	Elasticity of deposit demand	-275	Deposit spread
γ	Bank dividend payout rate	0.06	Capital/loans ratio
$\hat{N}$	Bank equity injection	0.016	Equity issuance/assets ratio
τ	Long term bond maturity	13.6	Bank assets maturity
D^*	Public service constraint threshold	6.93	2014 deposits/GDP ratio
κ^L	Leverage cost parameter	0.018	Elasticity of i_t^L to L_{SS}/N_{SS}

Central Bank: I set the persistence of the nominal rate ρ^{mp} to 0.93 and the Taylor rule coefficient ϕ^π to 2.74 following Coenen et al. (2019). $R^* = 0.46$ matches the average excess reserves/GDP ratio of the years 2014-2022, years when the policy rate was negative and reserves abundant. Finally, $\eta = 0.06$ replicates the two-tier system implemented by the ECB in 2019.

Table 4: Central bank parameters

Parameter	Description	Value	Target
ρ^{mp}	Taylor rule persistence	0.93	DSGE literature
ϕ^π	Taylor rule inflation coefficient	2.74	DSGE literature
R^*	Reserves target	0.46	Excess reserves/GDP ratio
η	Tiering coefficient	0.06	ECB tiering share

4 Results

My solution algorithm, implemented in Dynare and adapted from Abadi et al. (2023), finds the full nonlinear solution of the corresponding system of equations. This is crucial as the model is characterized by several non-linearities.

I solve the model under perfect foresight, following an unanticipated monetary policy shock at time $t = 0$. Specifically, at time $t = 0$, the central bank announces, by surprise, it will target nominal rate i^t for T periods. After that, it follows the Taylor rule. From time $t = 1$, the economy is fully deterministic and all agents have perfect foresight. The idea is simulating a forward guidance announcement.

I consider two different scenarios: an intermediate scenario where the central bank targets $i^t = -0.50\%$ for $T = 4$ periods, and an extreme scenario where the central bank targets $i^t = -1.00\%$ for $T = 4$ periods. In both scenarios, I solve the model with and without tiering and compare the results.

Figure 3 displays the impulse response function of the policy rate in steady state and after the two types of monetary policy shock. In steady state, the rate is constant at approximately 2%. After a monetary policy shock, the policy rate remains at the negative target for four periods and then quickly returns to positive values.

Figure 4 displays the entity of the transfer from the central bank to commercial banks under the two-tier system. The transfer is small in magnitude (0.05-0.10% of GDP), similarly to the actual tiering implemented by the ECB, and only lasts for the four periods when the interest rate is negative. However, as I show below, it has a significant impact on the economy.

Figure 5 displays the effect of the monetary policy shock on bank equity. Immediately after the announcement, banks obtain a capital gain since the price of long-term bonds increases. However, due to the public service constraint, banks obtain high losses on deposits, which, over time, erodes their equity. So much so that after two periods equity is lower than in steady state.

Finally, Figure 6 displays the leverage cost. Recall that leverage cost is increasing in loans and decreasing in bank equity. After the shock, firms demand more loans. Initially, the increase in bank equity more than compensates for the increase in loans, thus preventing

the leverage cost from increasing. However, as soon as equity gets eroded by losses on deposits, the leverage cost increases.

4.1 The effect of Tiering

To study the effect of tiering, I compare the IRFs of several key variables with and without tiering after the two types of monetary policy shocks. Specifically, in the following graphs, I plot the difference between the IRFs of the shocks in the model with tiering and without tiering. I interpret such a difference as the effect of tiering.

First, Figures 7 and 8 show the effects of tiering on equity and consequently the leverage rate. As expected, tiering, by supporting banks' profitability, increases (or allows for a lower decrease in) banks' equity. This, on the other hand, allows for a reduction in the leverage cost. Notice that the effect on the leverage cost is retarded, arising only when the banks have already lost all the capital gains they originally obtained after the shock.

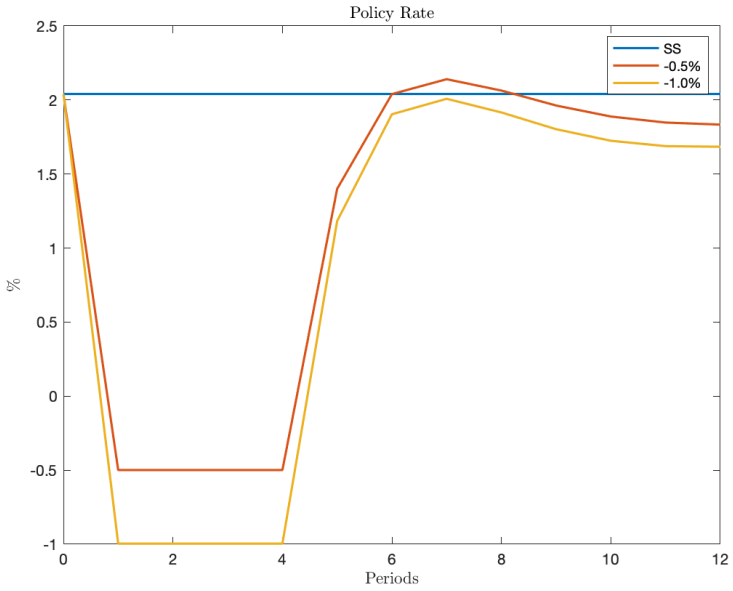
Figures 9 and 10 report the effect of tiering on lending and lending rates. By reducing the leverage cost, tiering increases lending in quantities and reduces the lending rate. Specifically, we observe an increase in lending of 0.75-0.15% and a decrease in the lending rate of 8-17 basis points. The effect is larger the more negative the policy rate becomes. This confirms that tiering has an expansionary effect.

Figure 11 shows that output increases about 0.01-0.02% thanks to tiering. The effect is small but recall that tiering is small in this model and that it can be implemented on a larger scale.

Figure 12, on the other hand, shows that consumption decreases up to 0.015% in the intermediate scenario and up to 0.40% in the extreme scenario. This is because, to finance tiering, the central bank incurs losses that are rebated to the government and therefore imply higher taxes for households. Thus, the effect of tiering on households' welfare is uncertain.

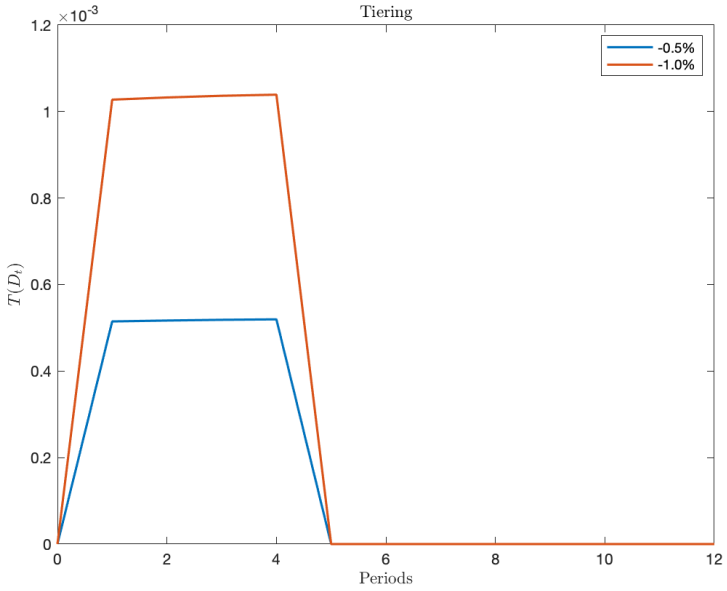
Finally, Figure 13, shows that tiering also helps the central bank increase inflation as it has an inflationary effect.

Figure 3: Policy rate



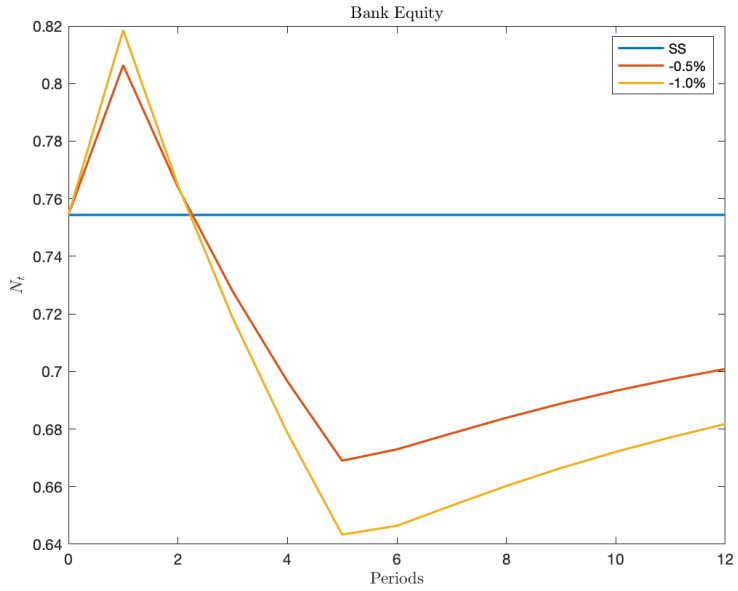
Notes: IRFs in the case with tiering.

Figure 4: Tiering



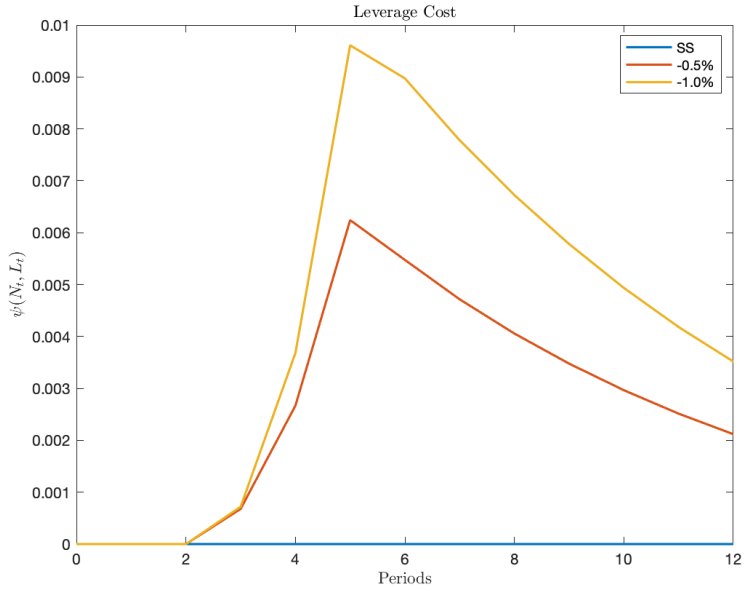
Notes: IRFs in the case with tiering.

Figure 5: Bank equity



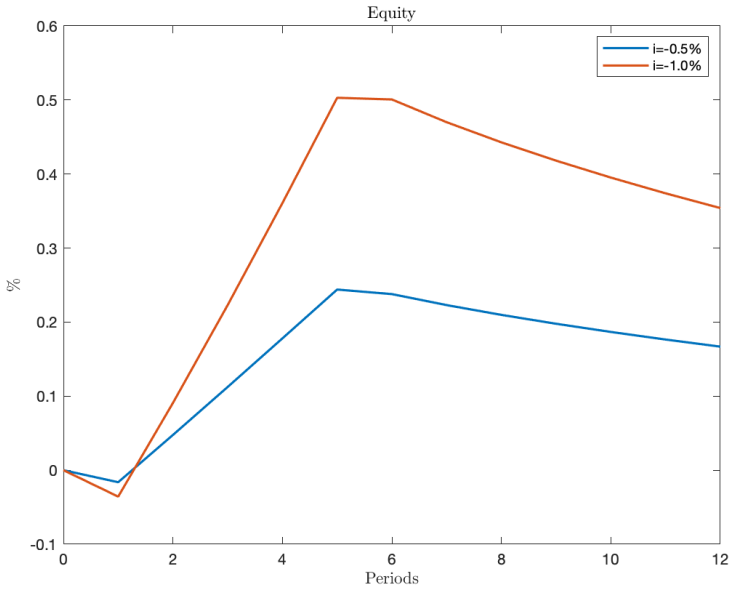
Notes: IRFs in the case with tiering.

Figure 6: Leverage cost



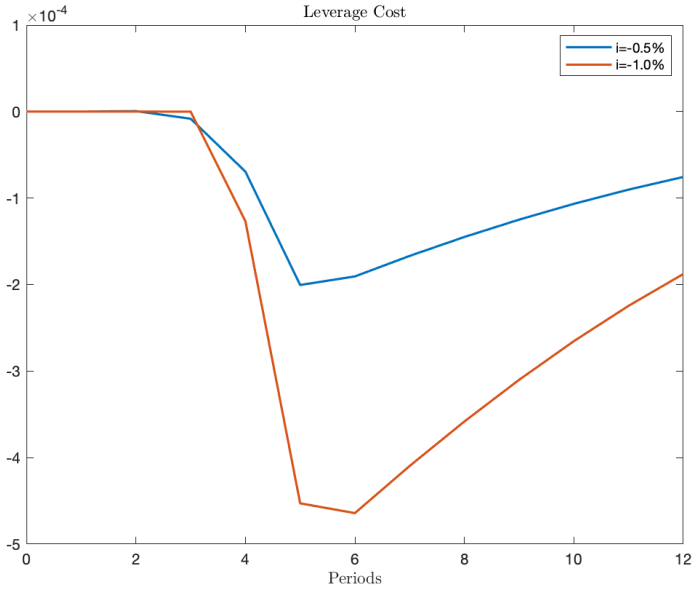
Notes: IRFs in the case with tiering.

Figure 7: Bank equity - Tiering



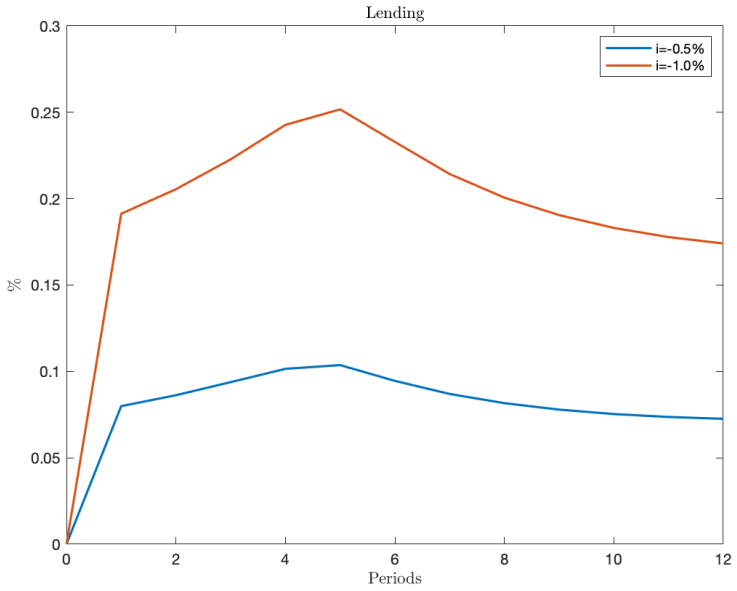
Notes: Difference between the IRF with tiering and the IRF without tiering.

Figure 8: Leverage cost - Tiering



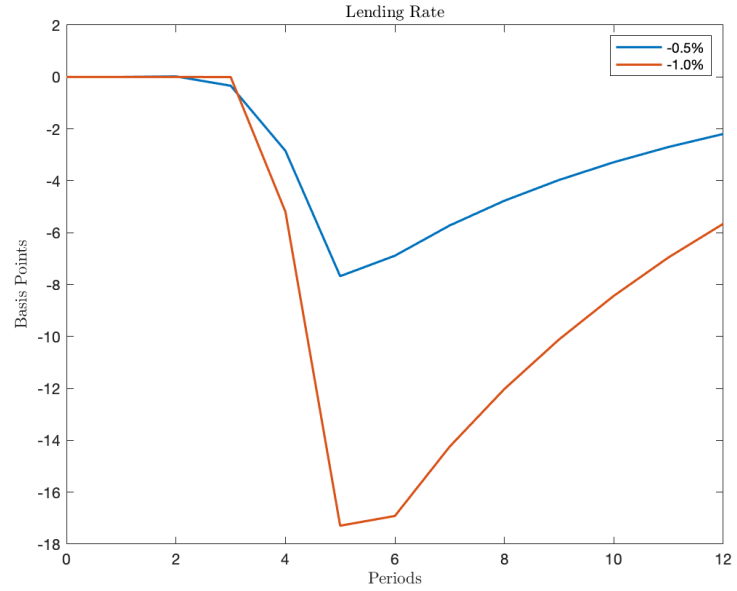
Notes: Difference between the IRF with tiering and the IRF without tiering.

Figure 9: Lending - Tiering



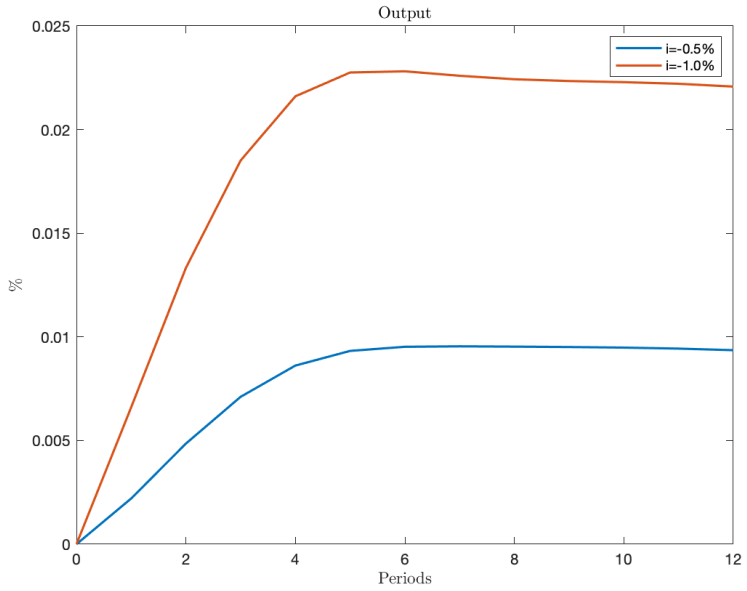
Notes: Difference between the IRF with tiering and the IRF without tiering.

Figure 10: Lending Rate - Tiering



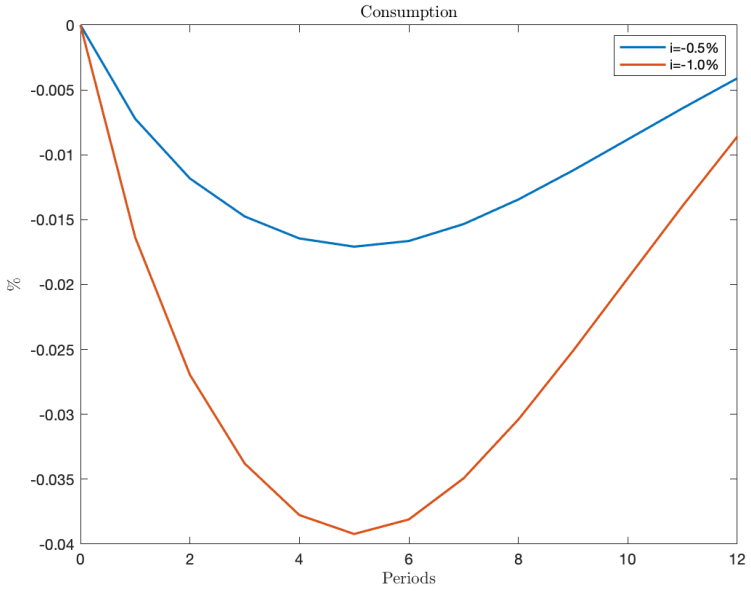
Notes: Difference between the IRF with tiering and the IRF without tiering.

Figure 11: Output - Tiering



Notes: Difference between the IRF with tiering and the IRF without tiering.

Figure 12: Consumption - Tiering



Notes: Difference between the IRF with tiering and the IRF without tiering.

5 Conclusion

This paper has examined the European Central Bank's two-tier system for remunerating banks' excess reserves, introduced in 2019 to mitigate the negative effects of prolonged negative policy rates. Through the development and analysis of a Neo-Keynesian macroeconomic model, we have explored the implications of this tiering system on the transmission of monetary policy.

The findings demonstrate that tiering can effectively support the transmission of expansionary monetary policy. By increasing banks' equity and relaxing capital constraints, tiering enhances banks' ability to lend, thus promoting higher economic output and inflation rates. However, this comes at a cost, as the mechanism reduces household consumption due to the higher taxes required to finance the central bank's losses.

Our results highlight the dual role of tiering: while it serves as an effective tool to stimulate the economy under negative policy rates, it also imposes a trade-off by impacting household welfare. This trade-off underscores the need for careful consideration in the design and implementation of such policies.

Moreover, while the scale of tiering in this model is relatively small, its potential impact on a larger scale warrants further investigation.

Future research should aim to solve the full stochastic dynamic version of the model and calibrate it using detailed Eurozone data to provide a more comprehensive assessment. Additionally, exploring the long-term implications of maintaining tiering as a permanent fixture in monetary policy could offer valuable insights for central bank strategies in a persistently low or negative interest rate environment.

Overall, this study contributes to the understanding of unconventional monetary policy tools and their role in modern central banking, providing a theoretical foundation for the observed empirical effects of tiering on the banking sector and the broader economy.

References

Abadi, Joseph, Markus Brunnermeier, and Yann Koby (Aug. 2023). "The Reversal Interest Rate". In: *American Economic Review* 113.8, pp. 2084–2120.

Altavilla, Carlo, Miguel Boucinha, Lorenzo Burlon, et al. (2022). "Money Markets and Bank Lending: Evidence from the Adoption of Tiering". In: *SSRN Electronic Journal*. ECB Working Paper Series.

Altavilla, Carlo, Miguel Boucinha, and José-Luis Peydró (2018). "Monetary policy and bank profitability in a low interest rate environment". In: *Economic Policy* 33.96, pp. 531–586.

Bernanke, Ben S, Mark Gertler, and Simon Gilchrist (1999). "The Financial Accelerator in a Quantitative Business Cycle Framework". en. In.

Boucinha, Miguel et al. (2022). "Two-Tier System for Remunerating Excess Reserve Holdings". In: *SSRN Electronic Journal*. ECB Occasional Paper Series.

Coenen, Günter et al. (2019). *The New Area-Wide Model II: an extended version of the ECB's micro-founded model for forecasting and policy analysis with a financial sector*. Working Paper Series 2200. European Central Bank.

Draghi, Mario and Luis de Guindos (2019). *ECB Press Conference 19 September 2019*.

Eisenschmidt, Jens, Julian Schumacher, and Sebastiaan Pool (2022). "Who benefits from the two tier system for reserve remuneration". In: *SSRN Electronic Journal*. ECB mimeo.

Feenstra, Robert C. (Mar. 1986). "Functional equivalence between liquidity costs and the utility of money". In: *Journal of Monetary Economics* 17.2, pp. 271–291.

Freriks, Jorien and Jan Kakes (July 2021). *Bank interest rate margins in a negative interest rate environment*. Working Papers 721. DNB.

Gerali, Andre et al. (2010). "Credit and Banking in a DSGE Model of the Euro Area". In: *Journal of Money, Credit and Banking* 42.s1, pp. 107–141.

Hoffmann, Peter et al. (2019). "Who Bears Interest Rate Risk?" In: *The Review of Financial Studies* 32.8, pp. 2921–2954.

Kiyotaki, Nobuhiro and John Moore (1997). "Credit Cycles". In: *Journal of Political Economy* 105.2, pp. 211–248.

Macroeconomic Assessment Group (2010). *Assessing the Macroeconomic Impact of the Transition to Stronger Capital and Liquidity Requirements*. Tech. rep. Financial Stability Board and the Basel Committee on Banking Supervision 2010.

Mendicino, Caterina et al. (2020). "Bank capital in the short and in the long run". en. In: *Journal of Monetary Economics* 115, pp. 64–79.

Poterba, James M and Julio J. Rotemberg (1986). *Money in the Utility Function: An Empirical Implementation*. Working Paper 1796. National Bureau of Economic Research.

Repullo, Rafael (2023). *Notes on Tiering*.

Rotemberg, Julio J. (1982). "Sticky Prices in the United States". In: *Journal of Political Economy* 90.6, pp. 1187–1211.

Smets, Frank and Raf Wouters (Sept. 2003). "An Estimated Dynamic Stochastic General Equilibrium Model of the Euro Area". In: *Journal of the European Economic Association* 1.5, pp. 1123–1175.

Appendix

A Derivations

A.1 Monopolistic competition in deposits and loans

Let:

$$D_t \equiv \left(\int_0^1 D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj \right)^{\frac{\epsilon^D}{\epsilon^D - 1}}, \qquad (1 + i_t^D) \equiv \left(\int_0^1 (1 + i_{j,t}^D)^{1-\epsilon^D} dj \right)^{\frac{1}{1-\epsilon^D}}, \tag{29}$$

be, respectively, the CES aggregator of deposits and deposit rates.

The households' first order condition with respect to deposits held at bank j is:

$$\frac{1}{D_t} \int_0^1 D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj D_{j,t}^{-\frac{1}{\epsilon^D}} = \Lambda_t (1 + i_{j,t}^D). \tag{30}$$

Dividing the FOCs for deposits held at any two different banks j and j':

$$\frac{D_{j,t}}{D_{j',t}} = \left(\frac{1 + i_{j,t}^D}{1 + i_{j',t}^D} \right)^{-\epsilon^D}. \tag{31}$$

Multiplying both sides for $D_{j',t}$, elevating to the power $\frac{\epsilon^D - 1}{\epsilon^D}$, integrating over j, and elevating to the power $\frac{\epsilon^D}{\epsilon^D - 1}$:

$$\begin{aligned}
D_{j,t} &= (1 + i_{j,t}^D)^{-\epsilon^D} (1 + i_{j',t}^D)^{\epsilon^D} D_{j',t} \\
D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} &= (1 + i_{j,t}^D)^{1-\epsilon^D} (1 + i_{j',t}^D)^{\epsilon^D - 1} D_{j',t}^{\frac{\epsilon^D - 1}{\epsilon^D}} \\
\int_0^1 D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj &= \int_0^1 (1 + i_{j,t}^D)^{1-\epsilon^D} dj (1 + i_{j',t}^D)^{\epsilon^D - 1} D_{j',t}^{\frac{\epsilon^D - 1}{\epsilon^D}} \\
\left(\int_0^1 D_{j,t}^{\frac{\epsilon^D - 1}{\epsilon^D}} dj \right)^{\frac{\epsilon^D}{\epsilon^D - 1}} &= \left(\int_0^1 (1 + i_{j,t}^D)^{1-\epsilon^D} dj \right)^{\frac{\epsilon^D}{\epsilon^D - 1}} (1 + i_{j',t}^D)^{\epsilon^D} D_{j',t} \\
D_t &= (1 + i_t^D)^{-\epsilon^D} (1 + i_{j',t}^D)^{\epsilon^D} D_{j',t}.
\end{aligned}$$

Solving for $D_{j',t}$ (and suppressing the superscript '), we obtain demand for deposits:

$$D_{j,t} = \left(\frac{1+i_{j,t}^D}{1+i_t^D}\right)^{-\epsilon^D} D_t. \tag{32}$$

Banks' decision for $i_{j,t}^D$ is fully static. Internalizing demand for deposits, and assuming the public service constraint does not bind, bank j maximizes:

$$\max_{i_{j,t}^D} (i_t - i_{j,t}^D) \left(\frac{1+i_{j,t}^D}{1+i_t^D}\right)^{-\epsilon^D}. \tag{33}$$

The corresponding FOC is:

$$-\left(\frac{1+i_{j,t}^D}{1+i_t^D}\right)^{-\epsilon^D} - (i_t - i_{j,t}^D)\epsilon^D \left(\frac{1+i_{j,t}^D}{1+i_t^D}\right)^{-\epsilon^D-1} \frac{1}{1+i_t^D} = 0,$$

or, solving for $1+i_{j,t}^D$:

$$1+i_{j,t}^D = \frac{\epsilon^D}{\epsilon^D - 1}(1+i_t). \tag{34}$$

Otherwise, if the public service constraint binds, bank j simply sets $1+i_{j,t} = 1$.

Therefore, the full solution is:

$$1+i_{j,t}^D = \max\left\{\frac{\epsilon^D}{\epsilon^D-1}(1+i_t); 1\right\}. \tag{35}$$

Now let:

$$L_t \equiv \left(\int_0^1 L_{j,t}^{\frac{\epsilon^L-1}{\epsilon^L}} dj\right)^{\frac{\epsilon^L}{\epsilon^L-1}}, \qquad (1+i_t^L) \equiv \left(\int_0^1 (1+i_{j,t}^L)^{1-\epsilon^L} dj\right)^{\frac{1}{1-\epsilon^L}}, \tag{36}$$

be, respectively, the CES aggregator of loans and loans rates.

From the first order condition of bank-dependent intermediate goods producers, similarly

to what I did above with deposits, one can obtain demand for loans:

$$L_{j,t} = \left(\frac{1 + i^L_{j,t}}{1 + i^L_t} \right)^{-\epsilon^L} L_t. \tag{37}$$

Bank j maximizes:

$$\max_{i^L_{j,t}} (i^L_{j,t} - i_t - \psi(L_t, N_t)) \left(\frac{1 + i^L_{j,t}}{1 + i^L_t} \right)^{-\epsilon^L}, \tag{38}$$

by setting:

$$1 + i^L_{j,t} = \frac{\epsilon^L}{\epsilon^L - 1} \left(1 + i_t + \frac{\partial \Psi(N_{j,t}, L_{j,t})}{\partial L_{j,t}} \right). \tag{39}$$

A.2 Additional optimality conditions

A.2.1 Households

Households always discount at the policy rate i_t: When the policy rate is positive, they must be indifferent between investing in deposits (obtaining utility and rate i^D_t) and bonds (obtaining the higher rate i_t). On the other hand, when the policy rate is negative, banks do not accept enough deposits (D^* is lower than household savings, as I verify in the solution) so that households must invest their marginal savings in bonds. Therefore, the Euler condition is:

$$1 = \beta \frac{1 + i_t}{1 + \pi_{t+1}} \frac{\Lambda_{t+1}}{\Lambda_t}, \tag{40}$$

where:

$$\Lambda_t = (C_t - hC_{t-1})^{-\sigma} - \beta h (C_{t+1} - hC_t)^{-\sigma} \tag{41}$$

is the household marginal utility of consumption.

The FOC for labor supply is:

$$\chi H^\phi_t = \frac{W_t}{P_t} \Lambda_t. \tag{42}$$

While the FOC for deposit holdings is:

$$\zeta \frac{1}{D_t} = \beta \Lambda_{t+1} (1 + i^D_t) - \Lambda_t. \tag{43}$$

A.2.2 Intermediate goods producers

Let us define the user cost of capital $1 + r^K_{z,t}$ of intermediate goods producers of type z as:

$$1 + r^K_{z,t} = \frac{1 + i_{z,t}}{1 + \pi_{t+1}} Q^K_{z,t} - (1 - \delta) Q^K_{z,t+1}, \tag{44}$$

where $i_{b,t} = i^D_t$ and $i_{n,t} = i_t$.

The FOC of capital in intermediate goods producers of type z problem is:

$$K_{z,t} = \left(\frac{1 + r^K_{z,t}}{A_z \alpha \nu L^{(1-\alpha)\nu}_{z,t} P^X_t} \right)^{\frac{1}{\alpha\nu - 1}}. \tag{45}$$

While the FOC for labor is:

$$L_{z,t} = \left(\frac{W_t / P_t}{A_z (1 - \alpha) \nu K^{\alpha\nu}_{z,t} P^X_t} \right)^{\frac{1}{(1-\alpha)\nu - 1}}. \tag{46}$$

A.2.3 Retailers

Retailers set their prices $P_{k,t}$ according to the FOC:

$$\Lambda_t \left((1 - \epsilon) \left(\frac{P_{k,t}}{P_t} \right)^{-\epsilon} \frac{Y_t}{P_t} - \epsilon \frac{P^I}{P_t} \left(\frac{P_{k,t}}{P_t} \right)^{-(\epsilon+1)} Y_t - \theta \left(\frac{P_{k,t}}{P_{k,t-1}} - 1 \right) \frac{Y_t}{P_{k,t-1}} \right)$$
$$= -\beta \Lambda_{t+1} \theta \left(\frac{P_{k,t+1}}{P_{k,t}} - 1 \right) \frac{Y_t}{P^2_{k,t}}. \tag{47}$$

From this, following Abadi et al., 2023, one can derive the New Keynesian Phillips Curve:

$$\log(1 + \pi_t) = \frac{\epsilon - 1}{\theta} \log \frac{P^X_t}{P_t} + \beta \log(1 + \phi_{t+1}). \tag{48}$$

A.2.4 Capital goods producers

The FOC of capital goods producers of type z is:

$$\Lambda_t Q^K_{z,t}\left(1-\frac{\kappa^I}{2}\left(\frac{I_{z,t}}{I_{z,t-1}}-1\right)^2-\kappa^I\frac{I_{z,t}}{I_{z,t-1}}\left(\frac{I_{z,t}}{I_{z,t-1}}-1\right)\right)$$
$$=\Lambda_t-\beta\Lambda_{t+1}Q^K_{z,t+1}\kappa^I\left(\frac{I_{z,t+1}}{I_{z,t}}\right)^2\left(\frac{I_{z,t}}{I_{z,t-1}}-1\right) \quad (49)$$

A.2.5 Fisher equation

The real rate $1+r_t$ satisfies:

$$1+r_t=\frac{1+i_t}{1+\pi_{t+1}}. \quad (50)$$

A.3 Market clearing conditions

The aggregate resource constraint is:

$$Y_t=C_t+I_{b,t}+I_{n,t}, \quad (51)$$

where I have assumed the retail price adjustment cost, the investment rate adjustment cost, and the cost of leverage, are rebated to households so do not appear in the resource constraint.

The amount of loans must be equal to the total value of capital employed by bank-dependent intermediate goods producers:

$$L_t=Q^K_{b,t}K_{b,t+1} \quad (52)$$

And the total amount of corporate bonds must be equal to the total value of capital employed by non-bank dependent intermediate goods producers:

$$B^h_t+B^b_t=Q^K_{n,t}K_{n,t+1} \quad (53)$$

Notice that, while the total amount of bonds can be determined at all times, the amount

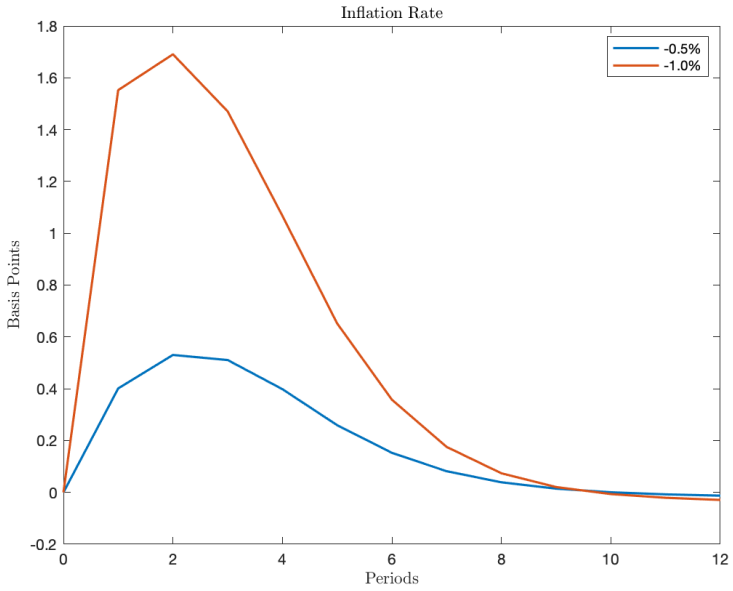
of bonds held by banks and households in steady state cannot be pinned down by the equations of the model. Therefore, I assume banks have zero bonds in time zero, and households own the whole of them as in Abadi et al. (2023).

The amount of government bonds held by banks and the central bank is pinned down by their respective balance sheet constraints:

$$G_t^b = D_t + N_t - L_t - R_t, \tag{54}$$

$$G_t^c = R_t. \tag{55}$$

Figure 13: Inflation - Tiering



Notes: Difference between the IRF with tiering and the IRF without tiering.